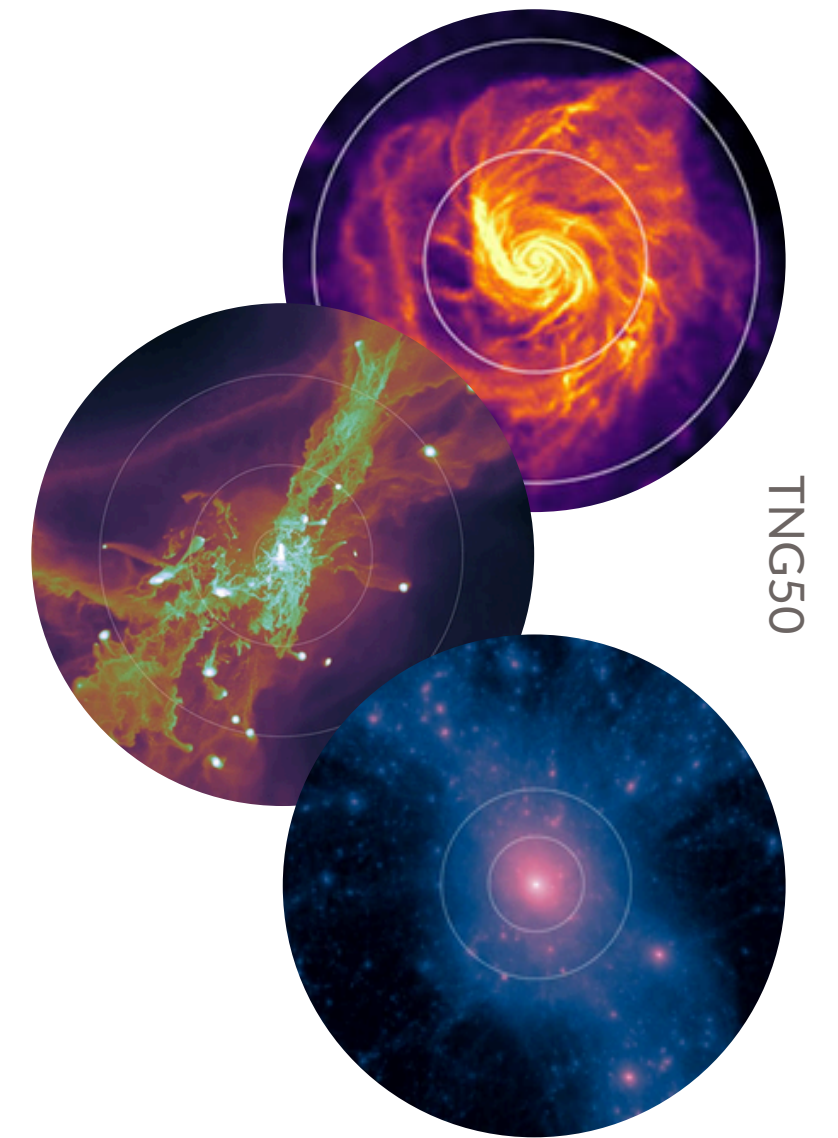


Spiral-Arm Morphology in IllustrisTNG50 Disk Galaxies and Its Relation to Dark Matter Halo Properties

Daniel Herald Grueso¹
Juan Carlos Muñoz-Cuartas²

FACom · Institute of Physics
Universidad de Antioquia
¹daniel.certuche@udea.edu.co
²juan.munozc@udea.edu.co

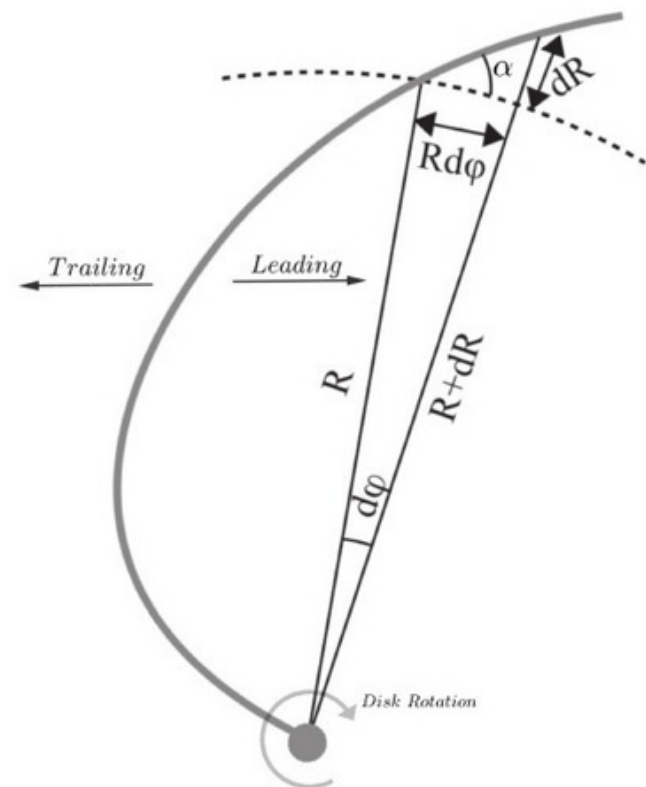


Spiral Arms in Disk Galaxies

Spiral-arm styles (Elmegreen et al., 1989; Yu & Ho, 2020):

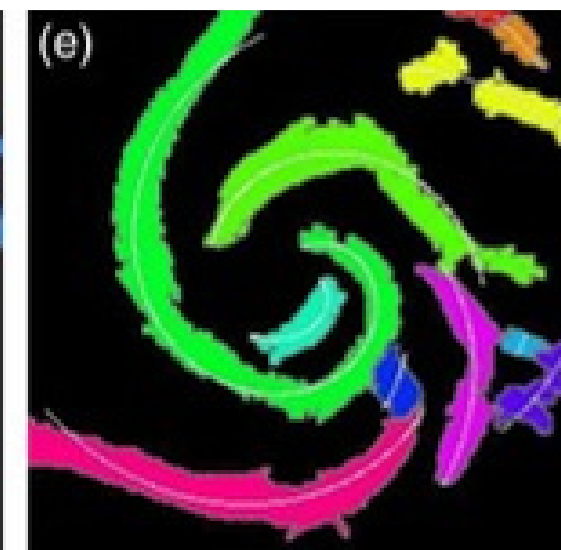
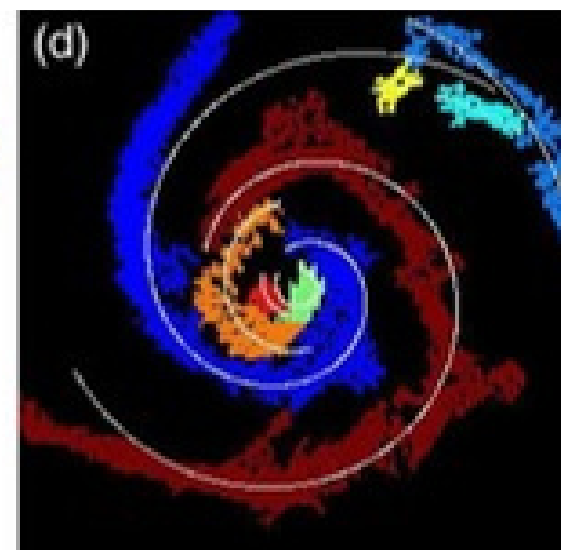
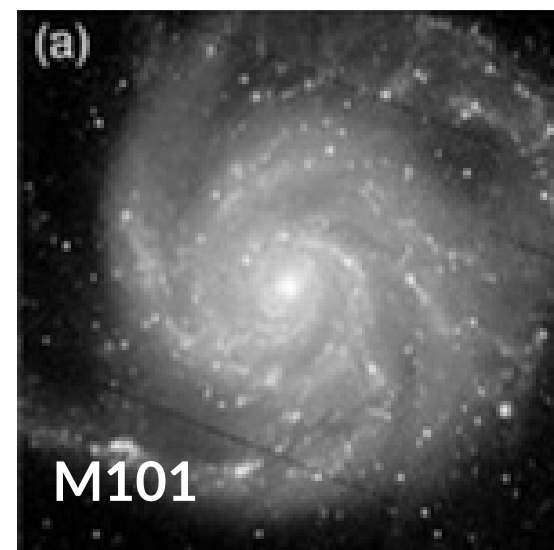
- ~10% grand design
- ~60% multi-armed
- ~30% flocculent

What are they? How do they form and move?



Pitch-Angle Definition

Observation



(Davis & Hayes, 2014)

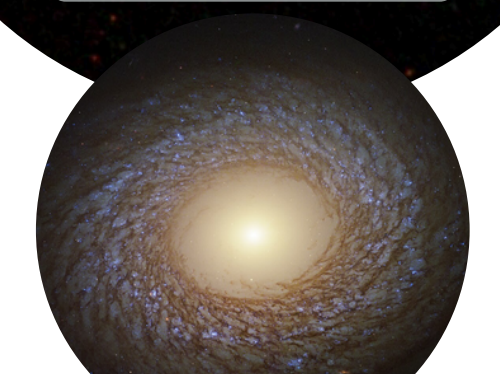
Defining an 'Arm' Is Difficult



NGC 1566 NASA/ESA



NGC 3464 SDSS



NGC 2775 NASA/ESA

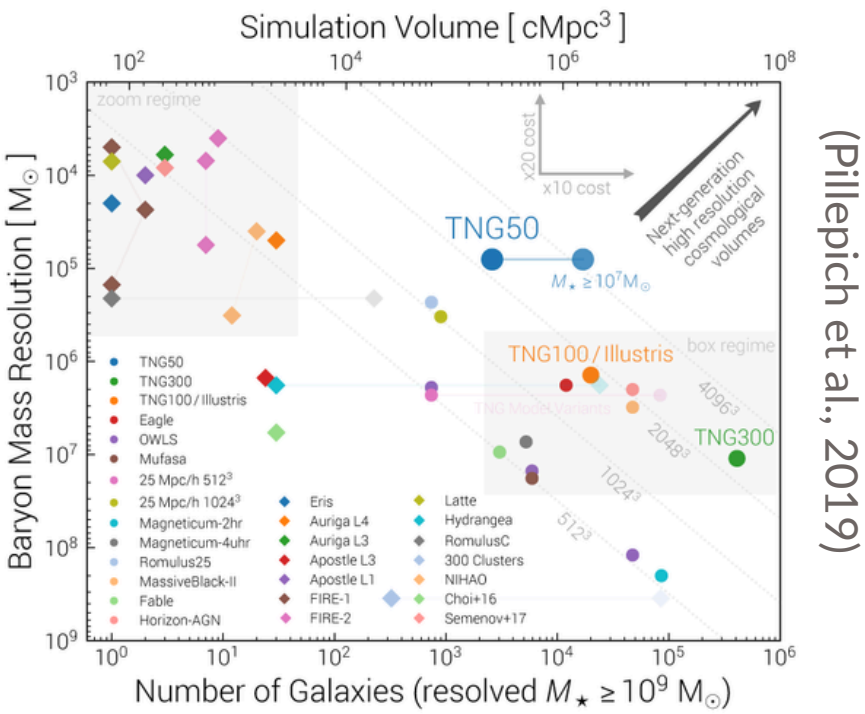
This context leads to a central question:

How do the properties of a galaxy's host dark matter halo relate to the morphology of its spiral arms?

- New patterns, new tests: disk–halo correlations can distinguish among density-wave, swing-amplification, and interaction scenarios.

What Sample Did We Use?

- TNG50
- 2.7 TB snapshot at $z = 0$
- Gas mass $> 10^{9.6} M_{\odot}$ (preselection)
- 377 subhalos total; 309 passed the filters



Parámetro	Variable
Coordenadas	x, y, z
Velocidad	vx, vy, vz
Momento angular	lxvel, lyvel, lzvel
Potencial gravitacional	Potential
Energía interna	U
Densidad	rho

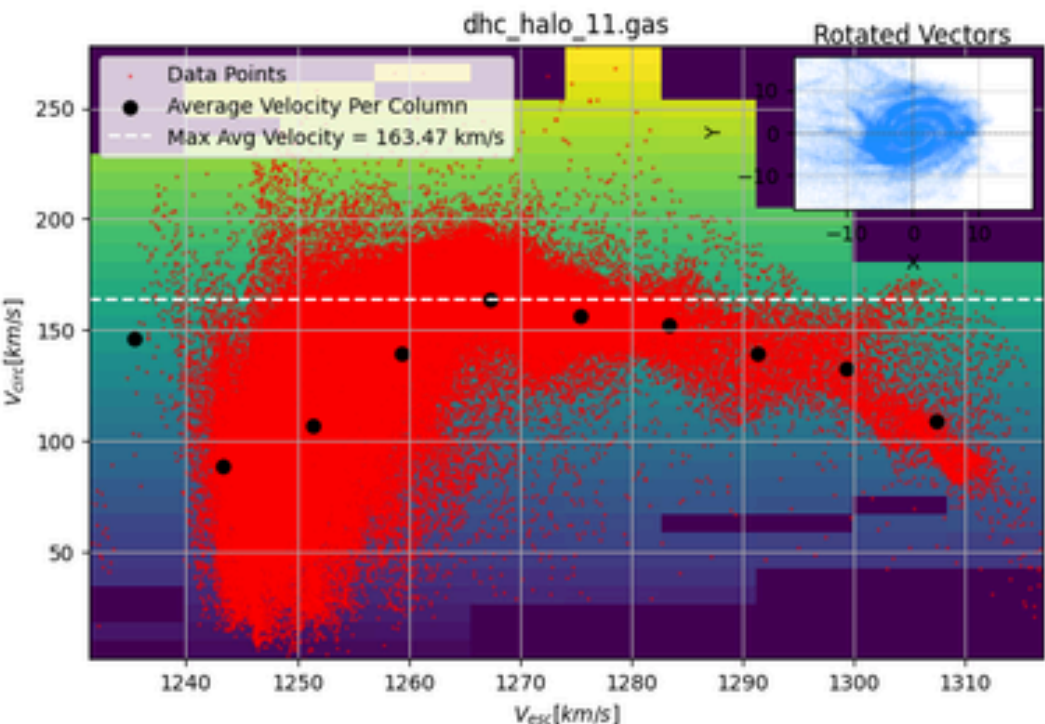
Table 1: Physical properties from IllustrisTNG gas cells (PartType0).

Parámetro	Variable
Masa total	logmass
Masa de gas	logmass0
Masa de materia oscura	logmass1
Masa estelar	logmass4
Radio	SubhaloHalfmassRad
Velocidad máxima	SubhaloVmax
Radio ($V_{circMax}$)	SubhaloVmaxRad

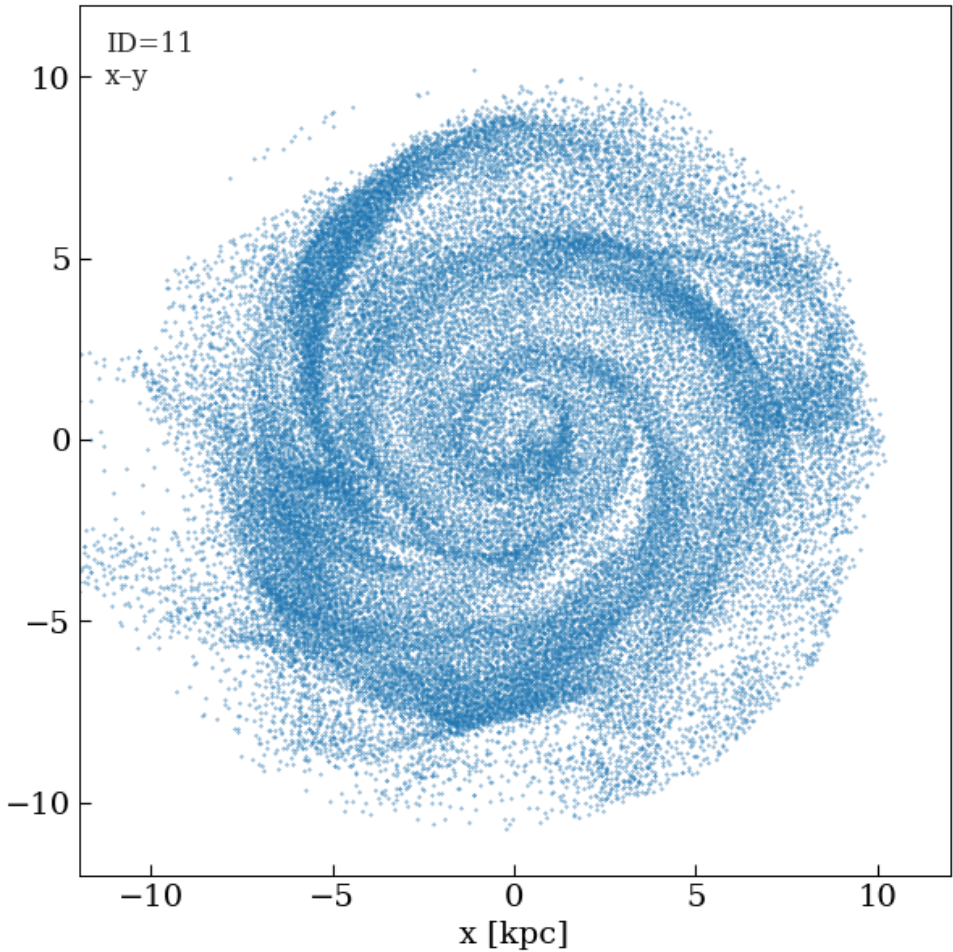
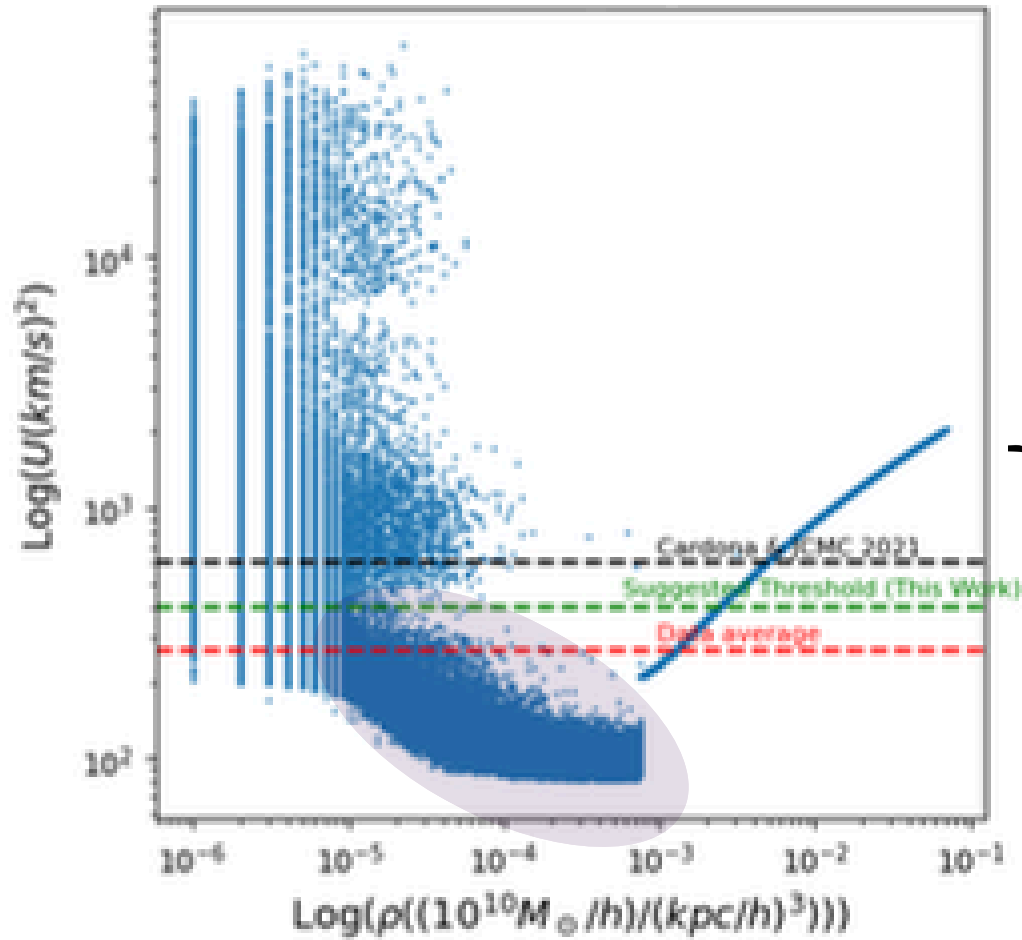
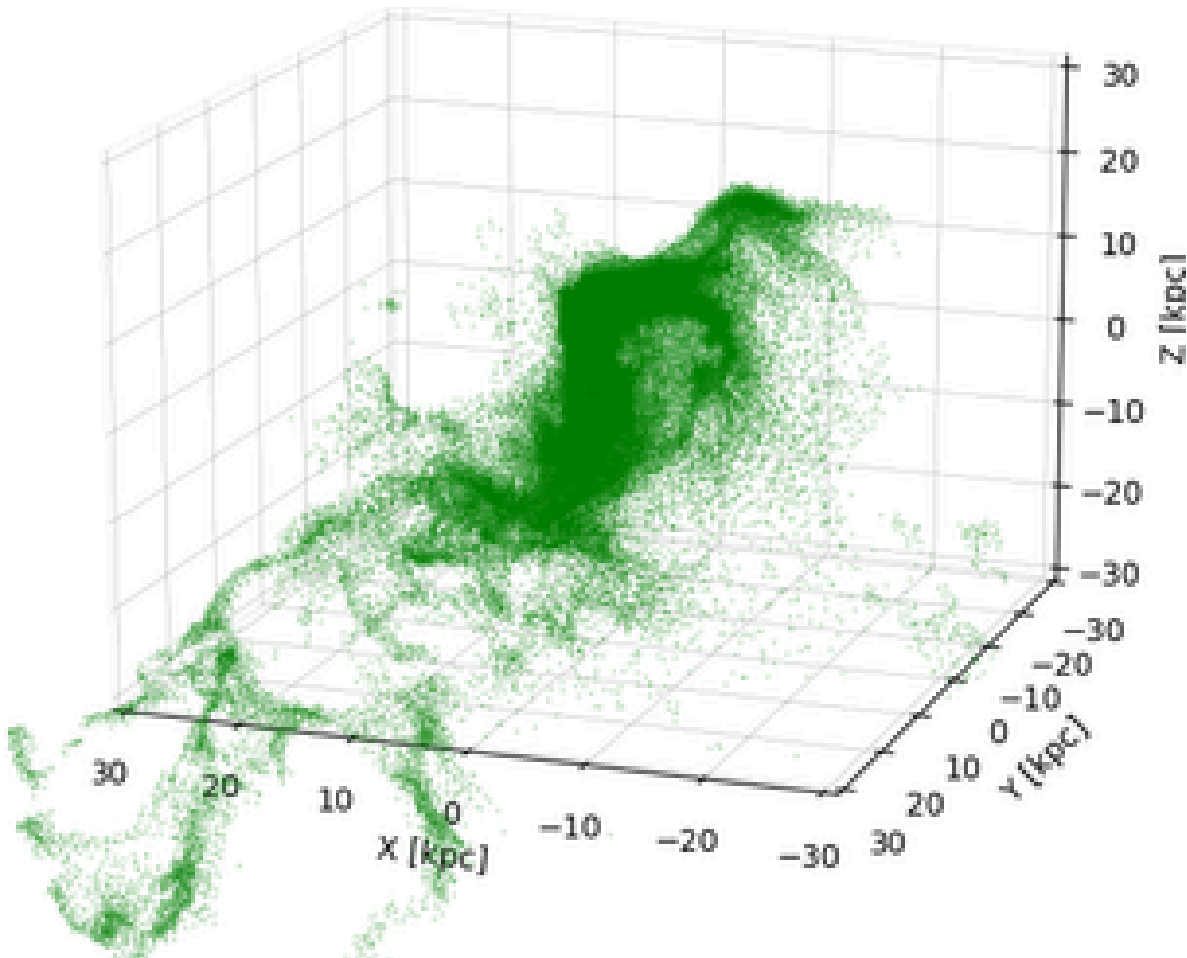
Table 2: Physical properties extracted for the subhalos.

Disk-Galaxy Selection

- ✓ Gas-Disk Inspection
 - ✓ Density-Based Spatial Clustering (DBSCAN)
 - ✓ Internal-Energy Selection to Isolate Disk Gas
 - ✓ Orbital-Circularity Selection
 - ✓ Face-On Deprojection
- 309 disks retained



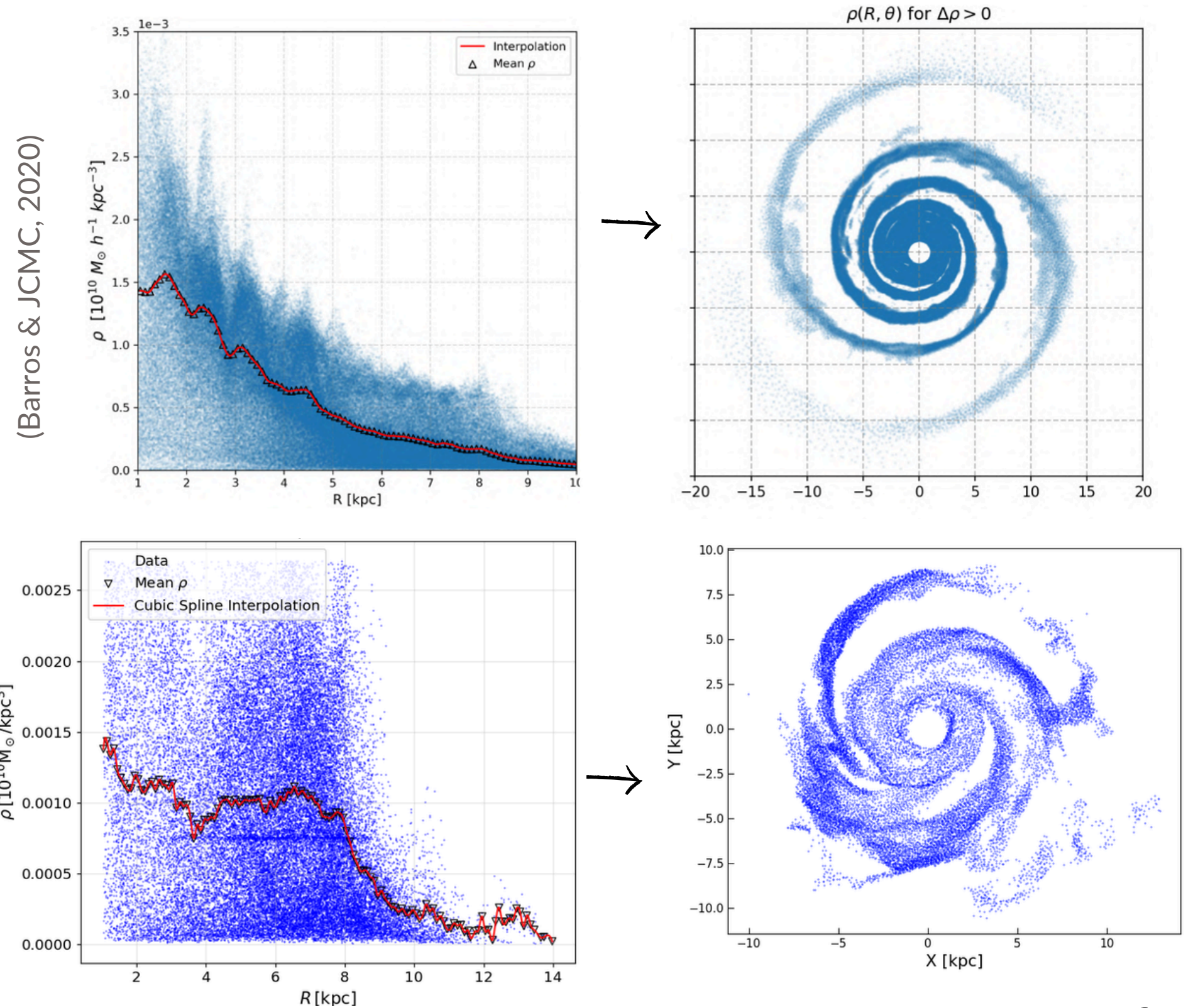
(Cardona & Muñoz-Cuartas, 2021)



Morphology Measurements

- ☑ Mean Density Profile
- ☑ Cubic Fit for $f(R)$
- ☑ Density Contrast

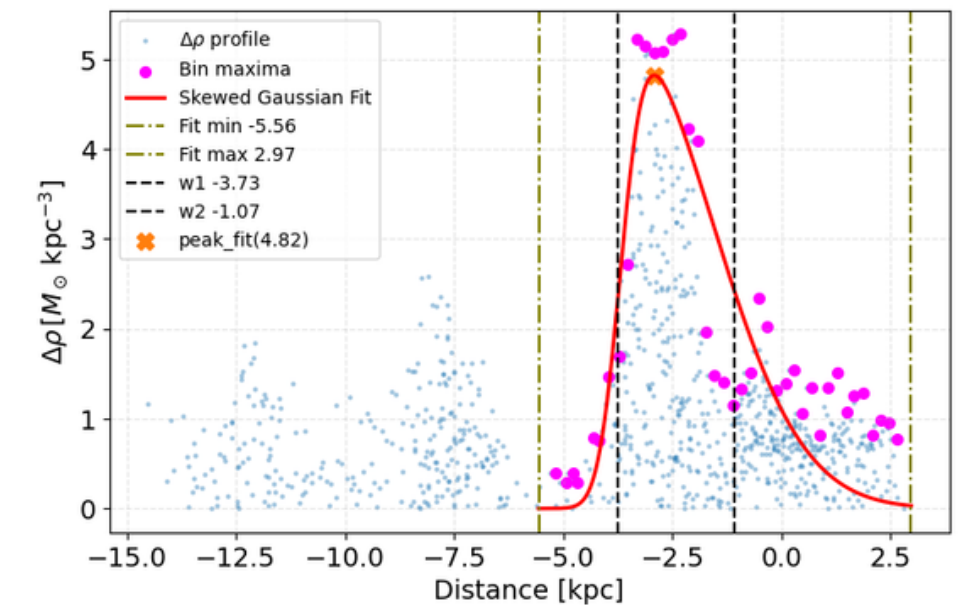
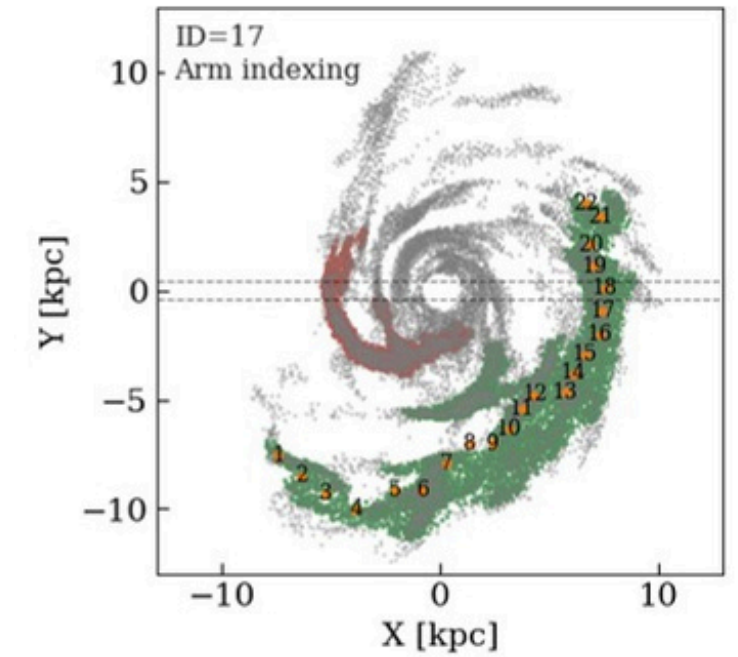
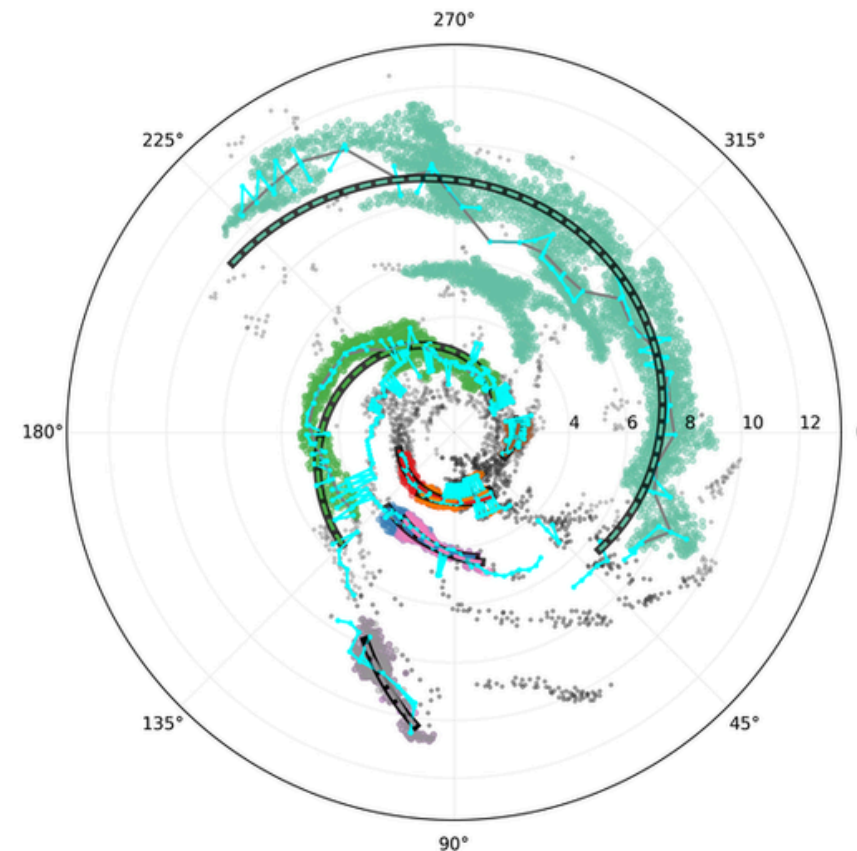
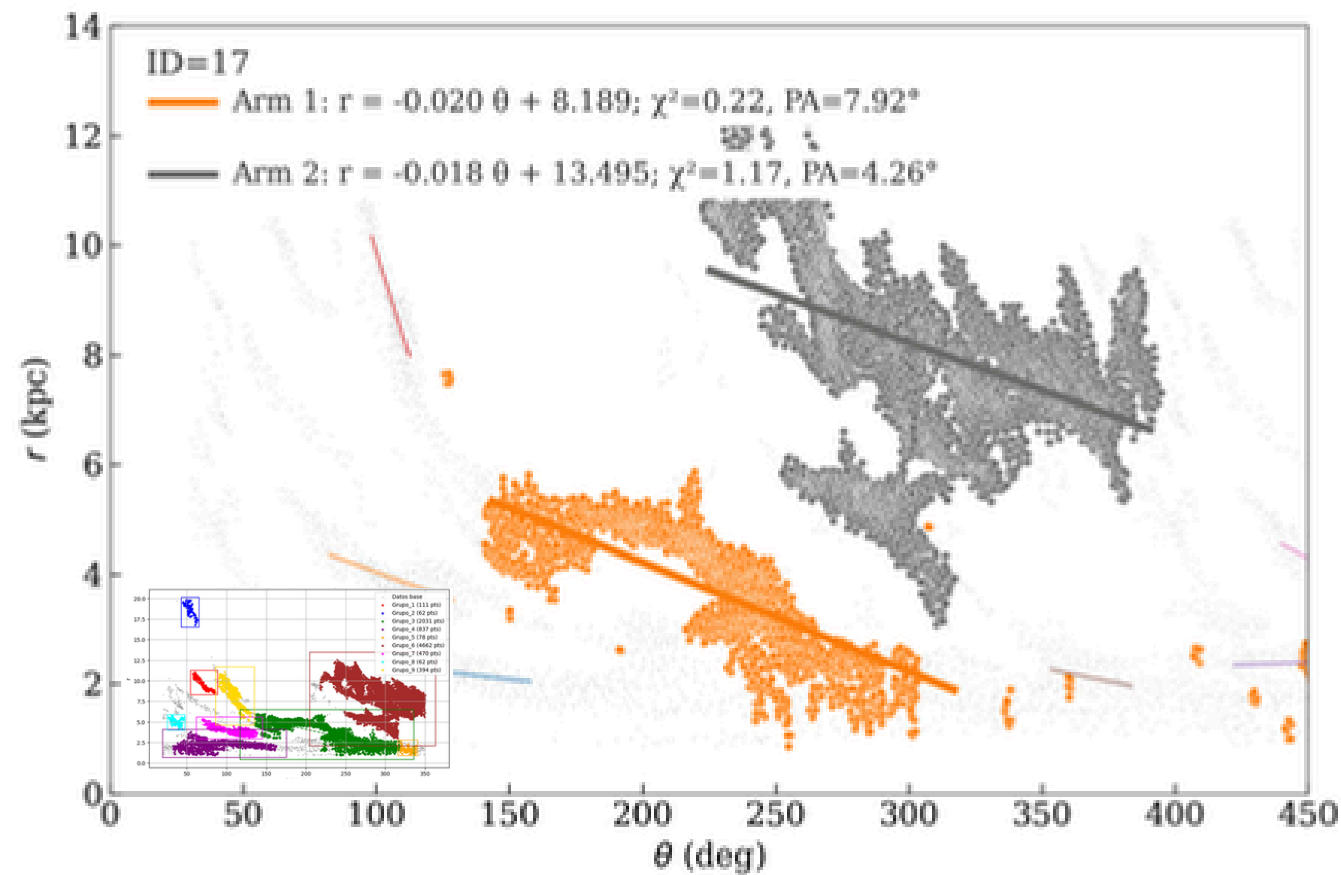
$$\Delta\rho = \frac{\rho(R, \theta, z)}{f(R)} - 1 \begin{cases} \Delta\rho < 0: \text{Background particles.} \\ \Delta\rho > 0: \text{Overdense regions.} \end{cases}$$



Morphology Measurements

$$r(\theta) = r_0 e^{k\theta} \longrightarrow r(\theta) \simeq r_0(1 + k\theta) = r_0 + r_0 k \theta \longrightarrow \text{PA} = \tan^{-1} k$$

- ☑ Points Tracing Arms in (r, θ)
- ☑ BFS Clustering and Polar-Plane Arm Fitting (PA)
- ☑ Arm-Center Extraction
- ☑ Width Measurement and Fitting

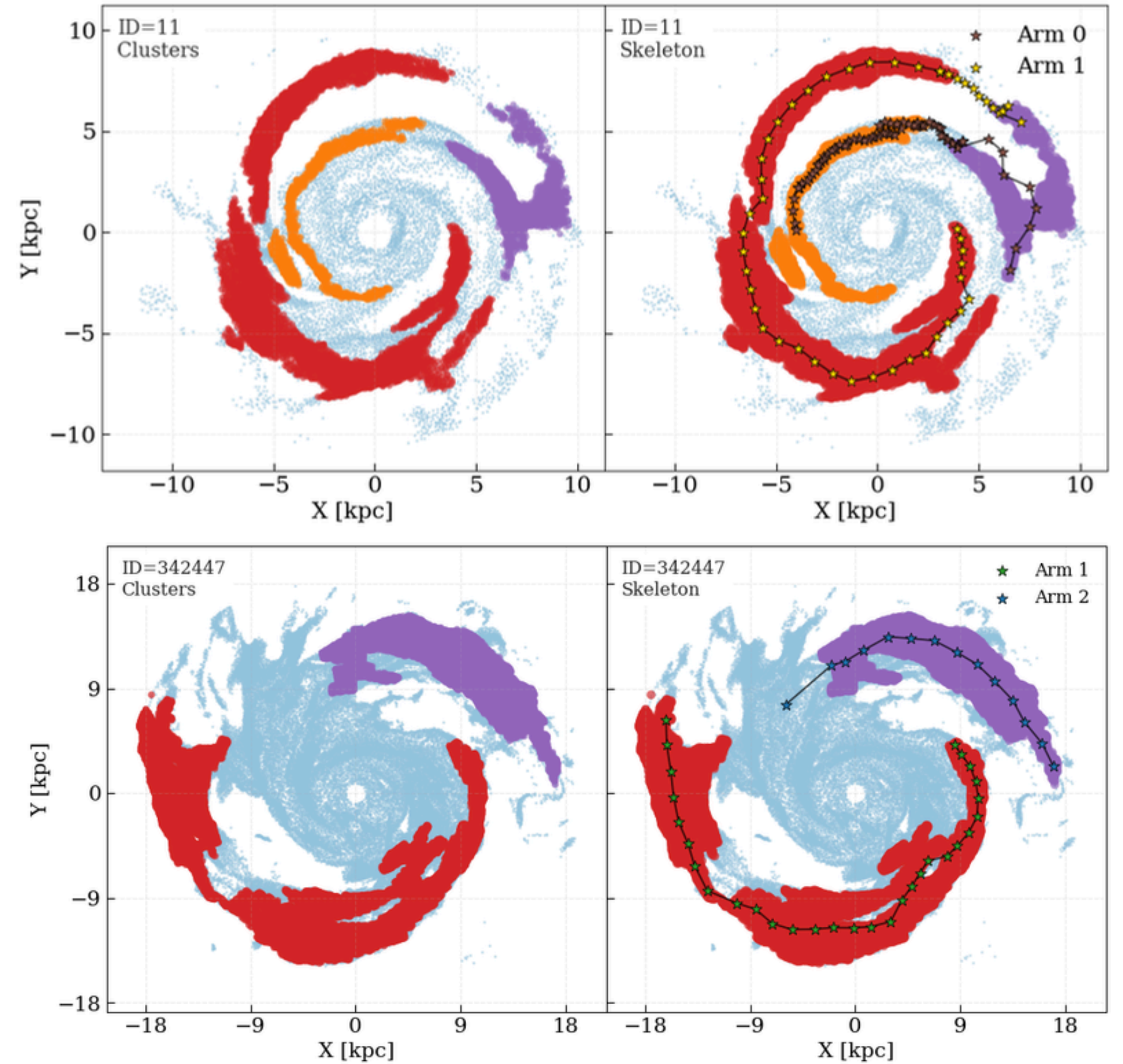
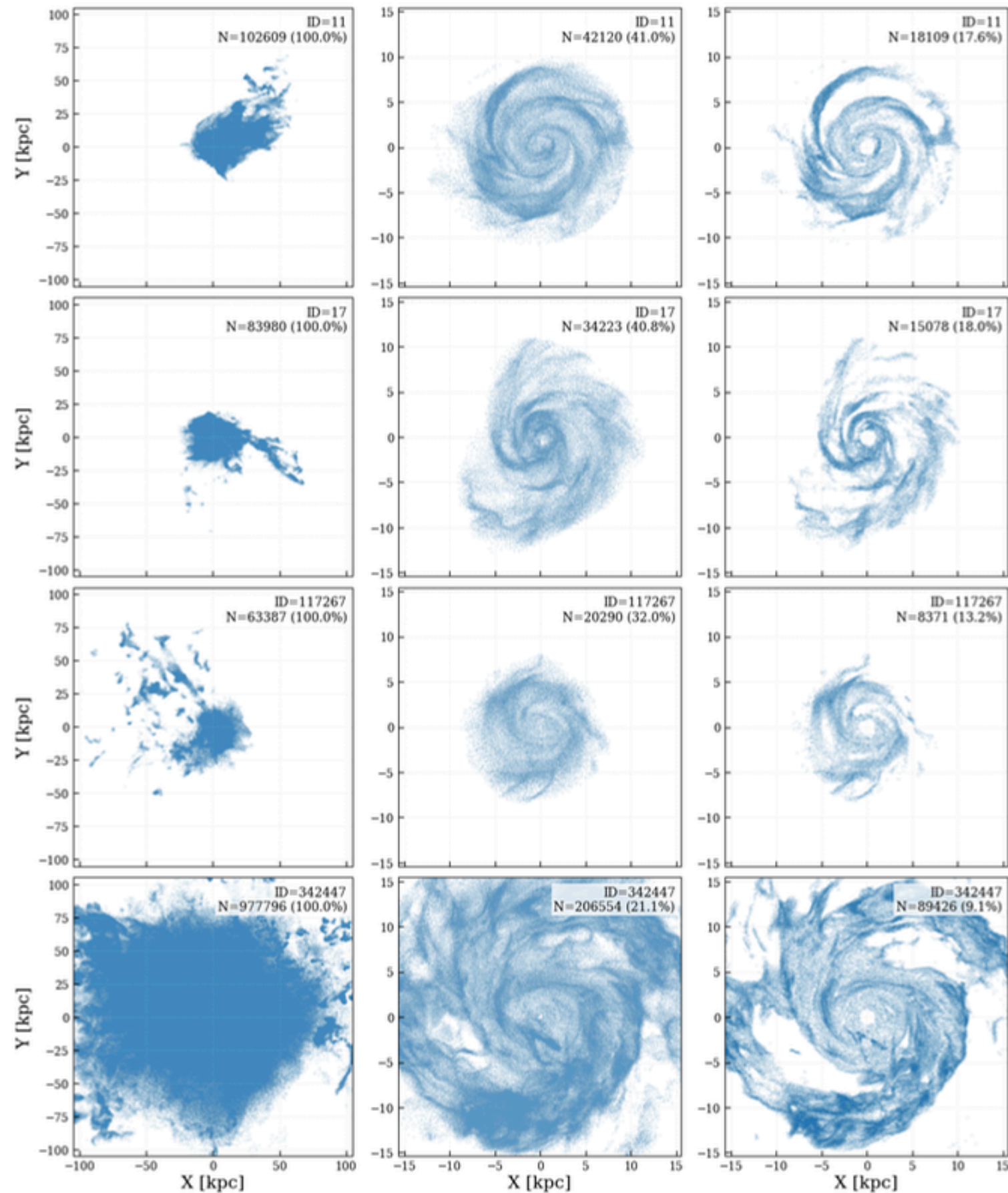


$$G(x) = \frac{A}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} \left[1 + \text{erf}\left(\gamma \frac{x-\mu}{\sqrt{2}\sigma}\right) \right]$$

Asymmetric Gaussian Equation

Results

Method Coverage: ~80%

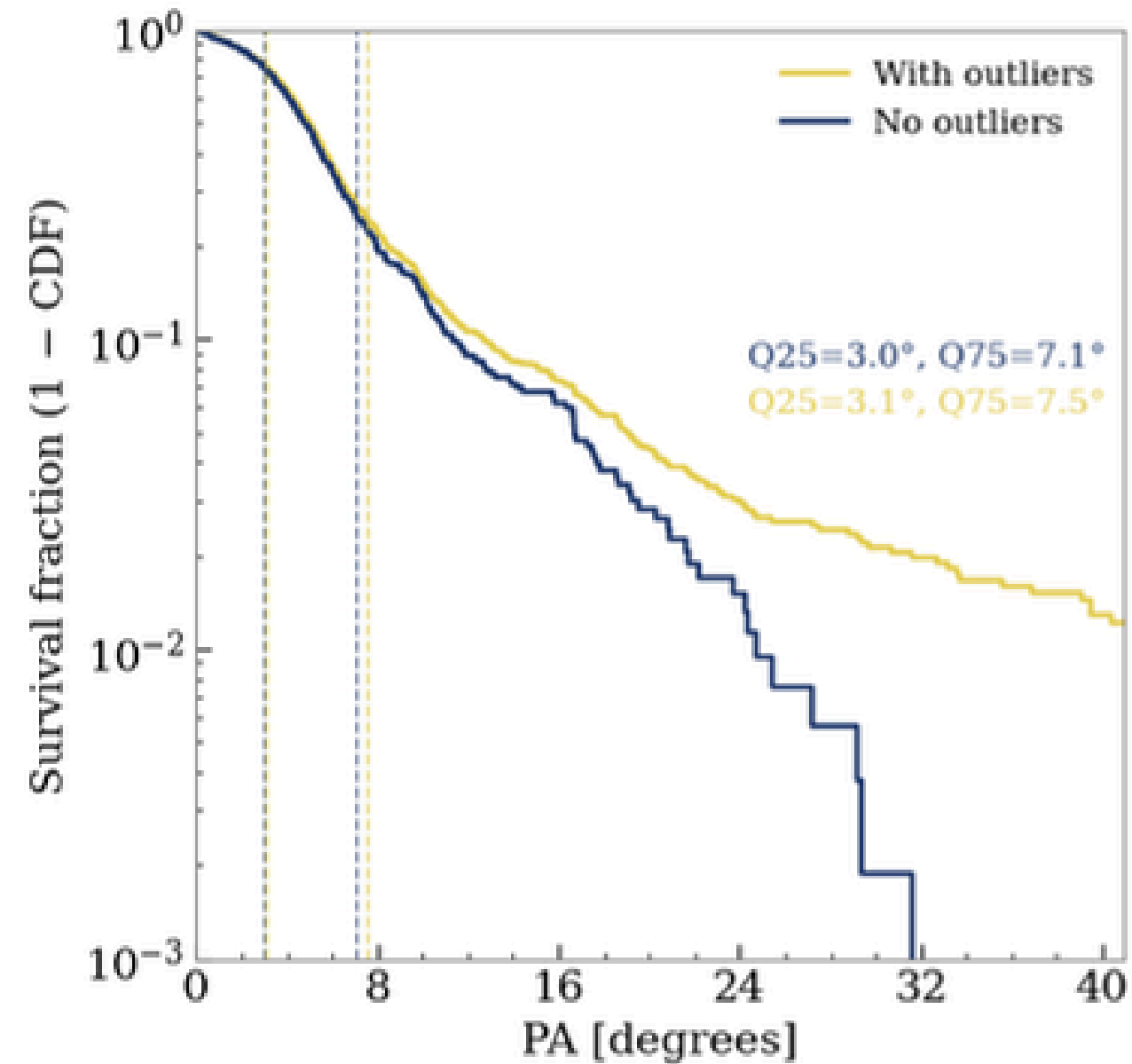


~80% yielded arms: pipeline coverage, not scientific validation.

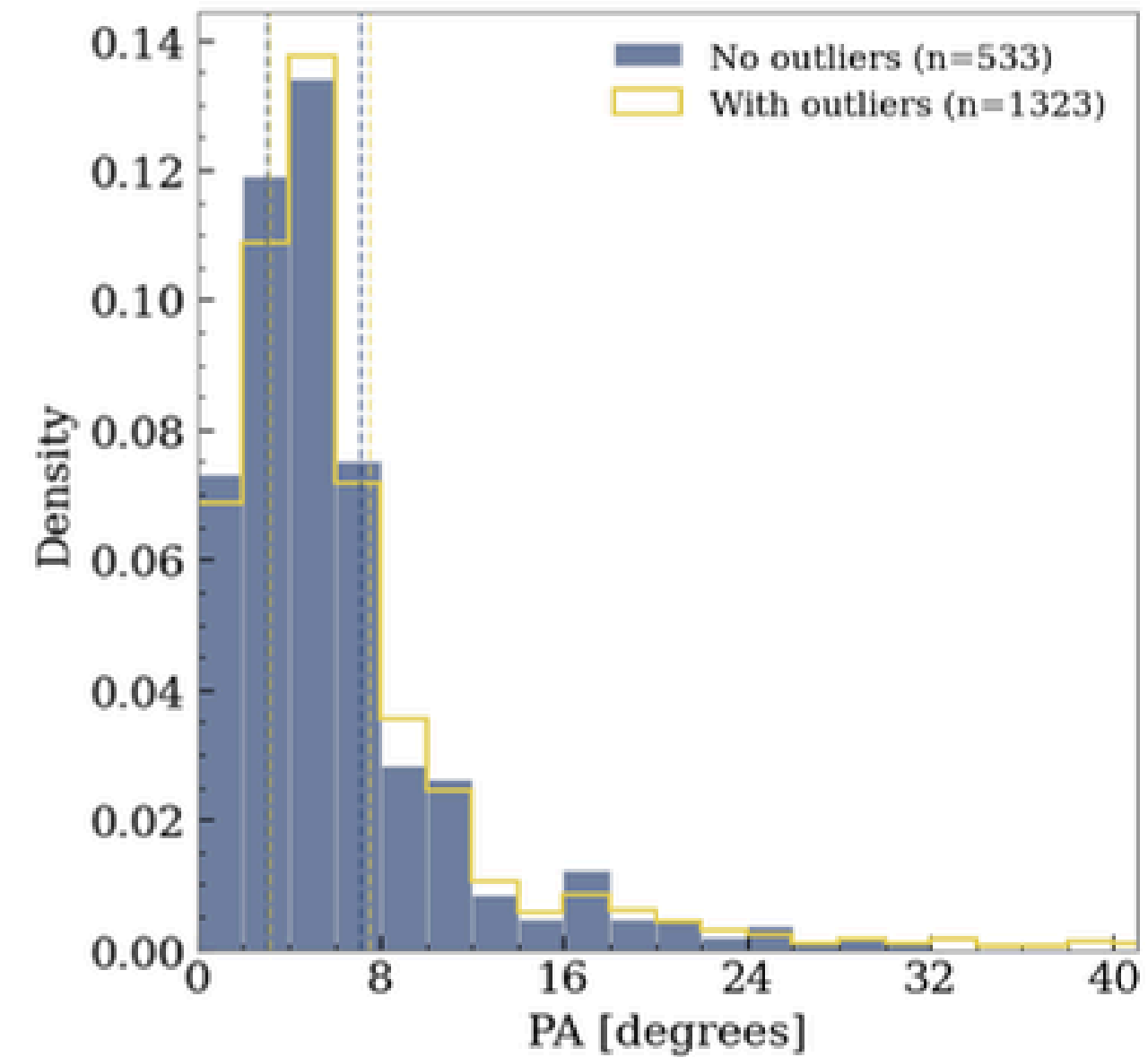
Results

Sample Filters

PA Survival Function



PA Density per Arm



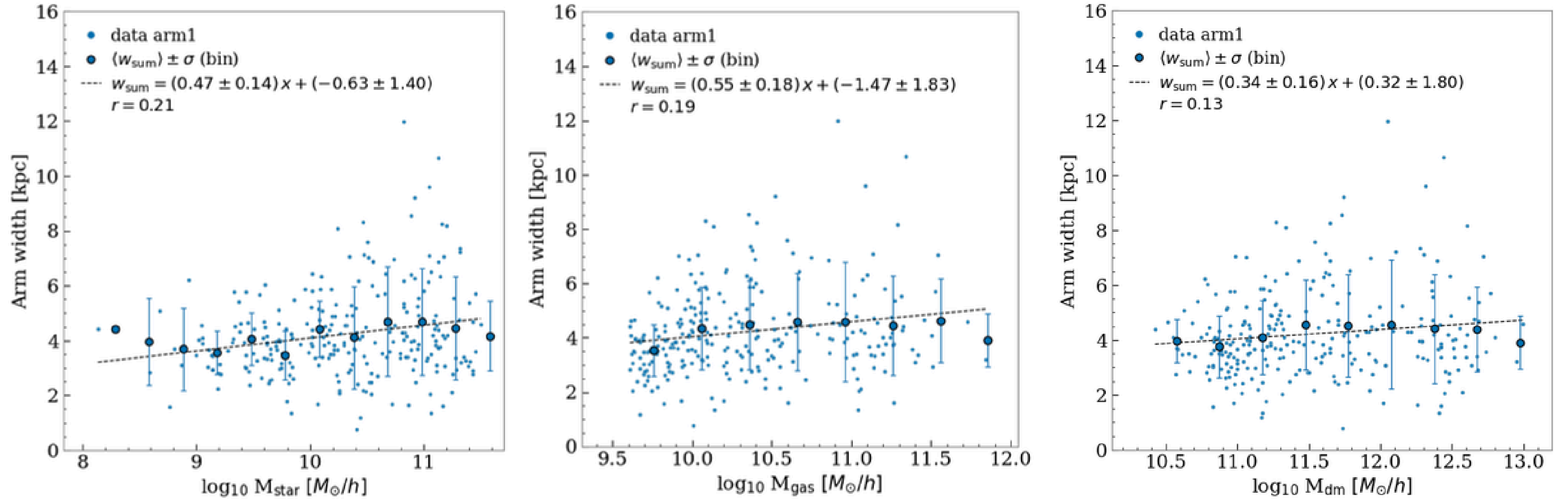
Samples: full $n = 1,323$ (309 disks); clean subset $n = 533$ (267 disks, two PRINCIPAL arms per galaxy)

Mean = 6.85°

Results

Halo Properties:

Width vs. Mass (3 Tracers)



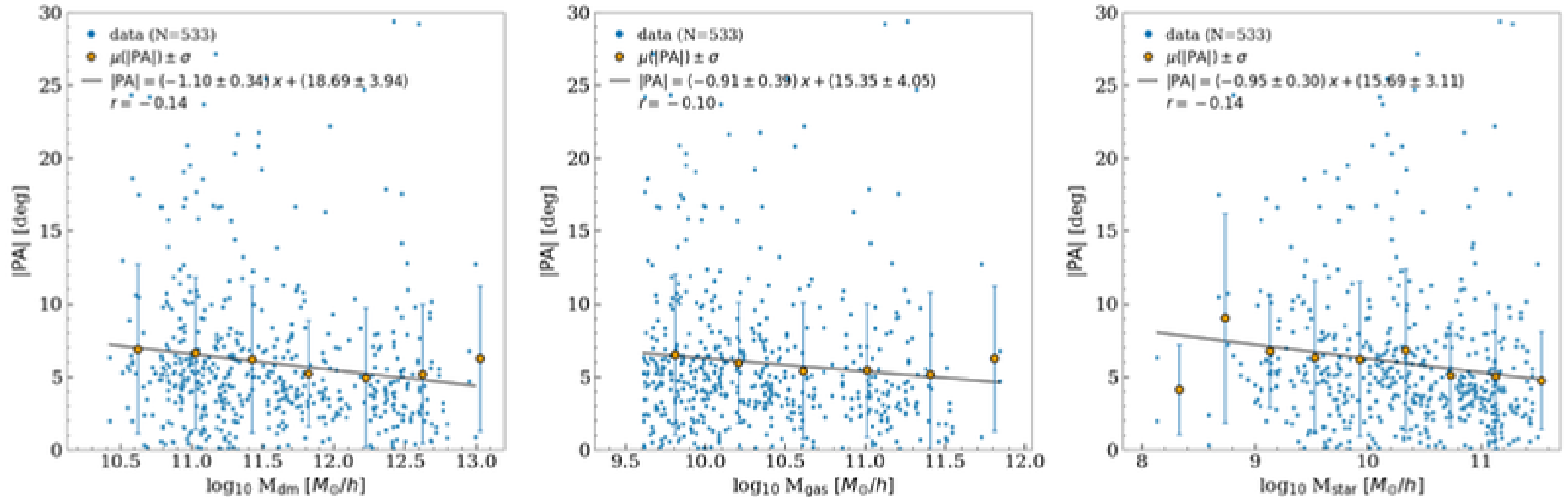
The original estimator suggests wider arms at higher stellar, gas, and dark matter mass.

The trend is tentative ($\leq 1.3\sigma$).

Results

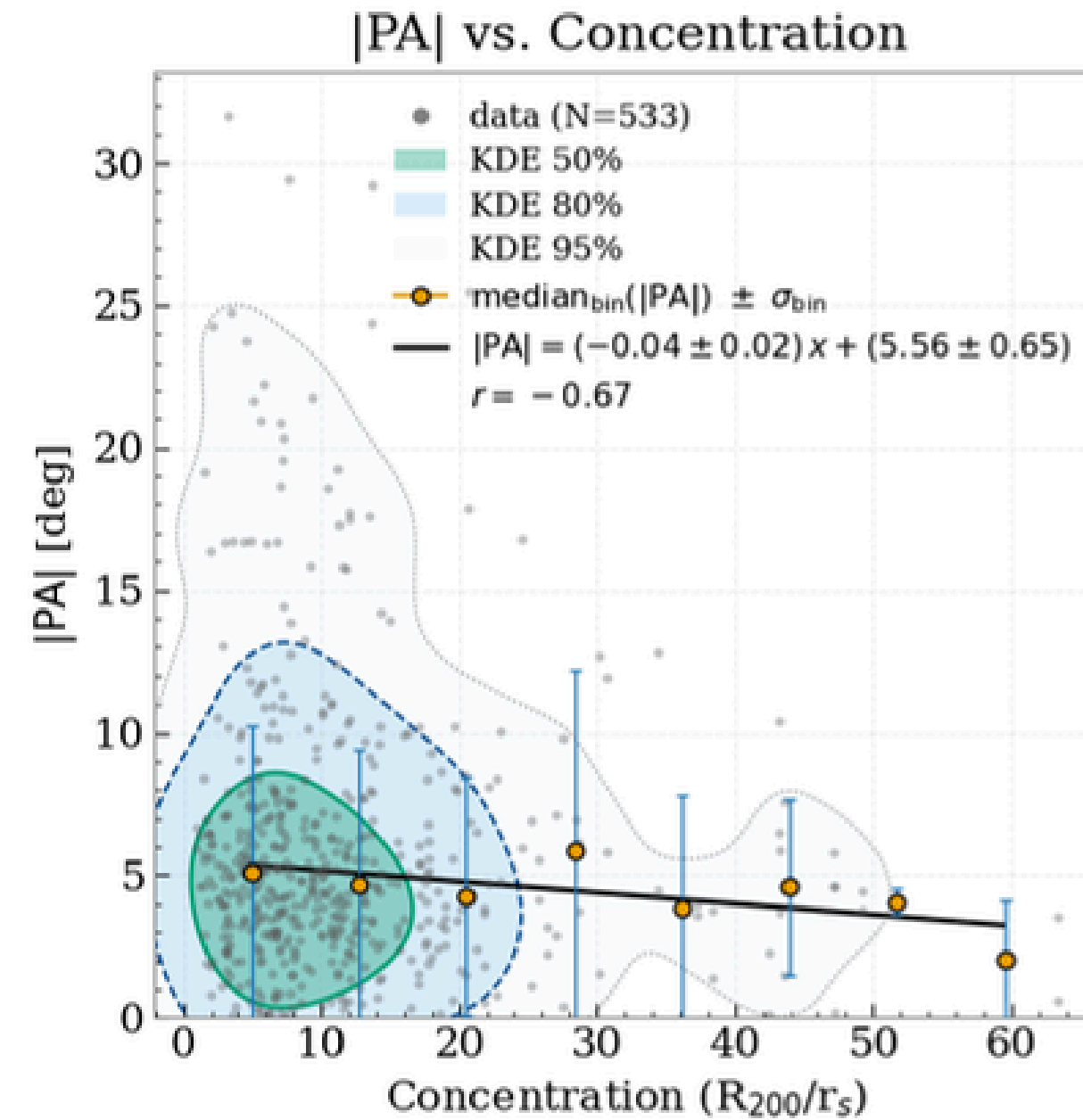
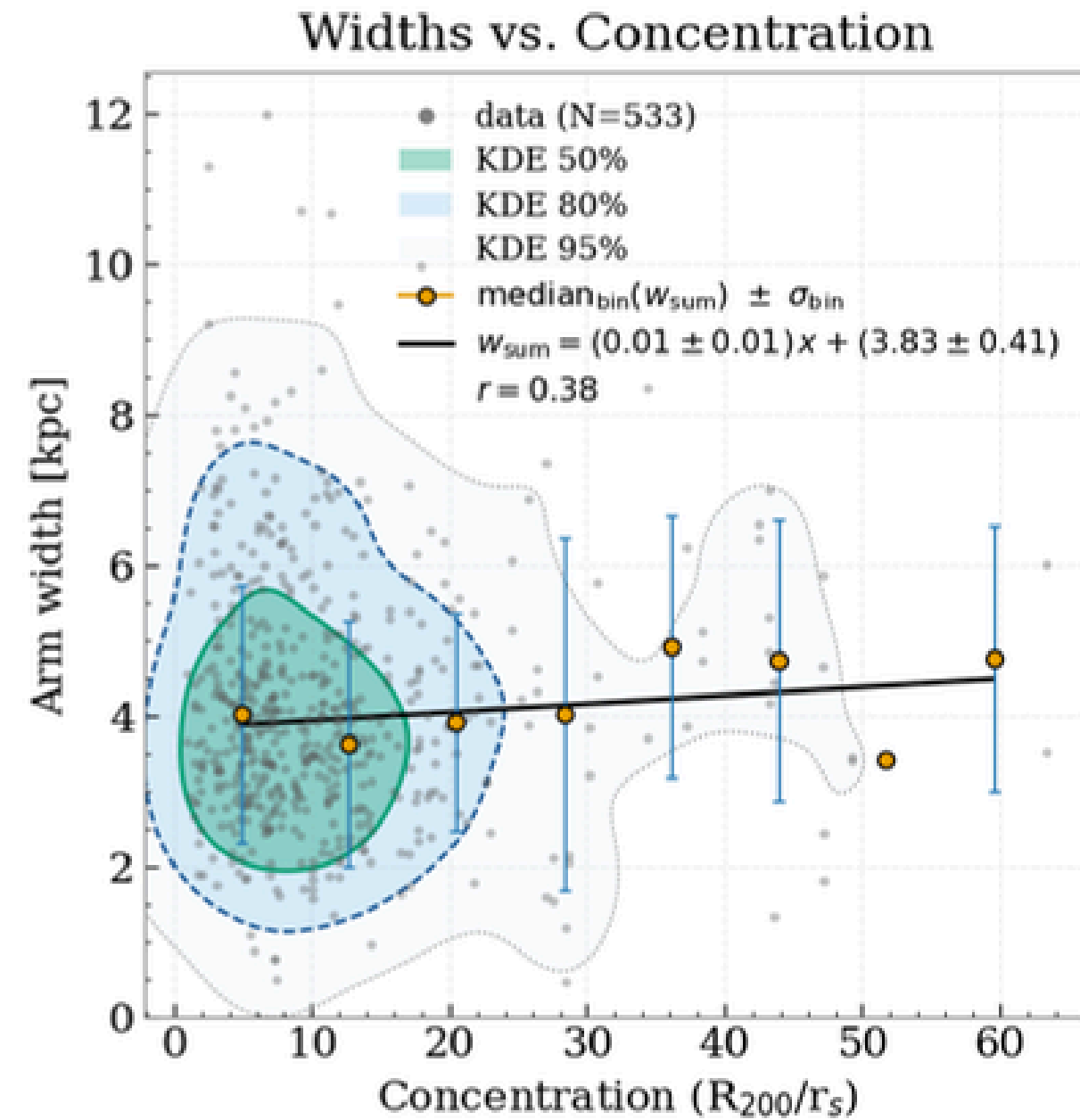
Halo Properties:

Pitch Angle vs. Mass (3 Tracers)

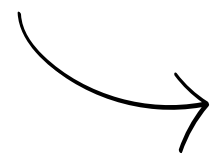


- |PA| decreases mildly with mass; tracer differences are not significant
- More massive systems may host slightly tighter arms, but the evidence is inconclusive.

Results



- Mean arm width is consistent with no concentration trend
- |PA| decreases mildly as concentration increases



Descriptive fit only; broader evidence is tentative and non-causal.

Takeaways

- We developed an astrophysical and numerical workflow that identifies galaxy disks and automatically extracts coherent spiral arms.
- **Arms were recovered in ~80% of filtered disks (method coverage).**
- With the original estimator, mean arm width increases modestly with halo, gas, and stellar mass. This tentative result needs testing with the revised estimator.
- No systematic amplitude–halo-mass trend was detected (Pearson $r \approx 0.06$ – 0.14).
- A weak pitch-angle–concentration anti-correlation appears in this sample. It does not establish that a stronger central potential causes tighter winding.
- Next: rerun revised widths across the full sample, enlarge the galaxy sample, quantify selection effects, and compare with observations.



Sources

Barros Ramirez, D. F. (2020). Star formation in disk galaxies and its relation with spiral structure in numerical simulations (Undergraduate thesis, Universidad de Antioquia).

Pillepich, A., et al. (2019). First results from the TNG50 simulation: the evolution of stellar and gaseous discs across cosmic time. *Monthly Notices of the Royal Astronomical Society*, 490(3), 3196–3233.

Silva-Villa, E., & Cano Gomez, X. (2022). NGC 5236's stars as tracers of arms and arm widths in spiral galaxies. *Monthly Notices of the Royal Astronomical Society: Letters*, 514(1), L22–L26.

Yu, S.-Y., & Ho, L. C. (2020). The statistical properties of spiral arms in nearby disk galaxies. *The Astrophysical Journal*, 900(2), 150.

Thank You

Questions and discussion
